# 5G-Enabled AI Smart Classroom Attendance System

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## Abstract

This project presents an AI-based smart attendance system integrated with 5G connectivity to automate classroom attendance. Using computer vision, it detects and recognizes students in real time, eliminating manual effort and proxy attendance. The system ensures high accuracy, low latency, and scalability, making it suitable for modern educational environments.

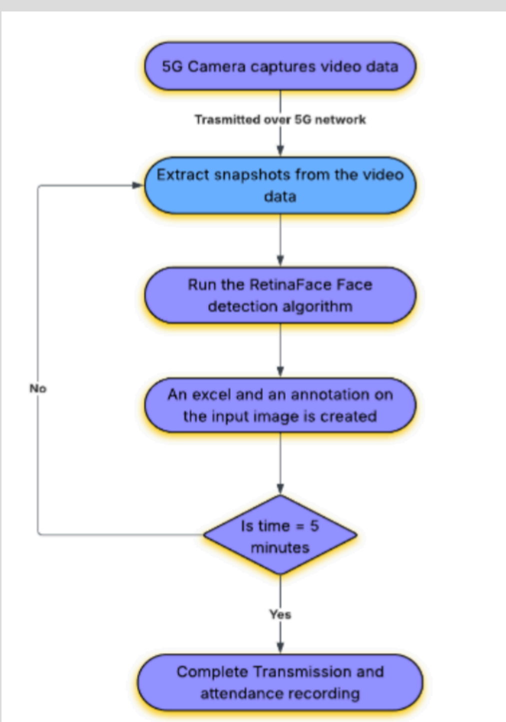
## Problem and Motivation

- Time-consuming and error-prone manually
- Proxy attendance is difficult to prevent
- Biometric/RFID systems require physical interaction
- Not scalable for large classrooms

**Need an automated and scalable solution using modern technologies like AI and 5G.**

## Proposed solution

We propose a real-time AI-based attendance system that captures classroom images using cameras, detects and recognizes faces automatically and marks attendance by matching with a database




Architecture flowchart

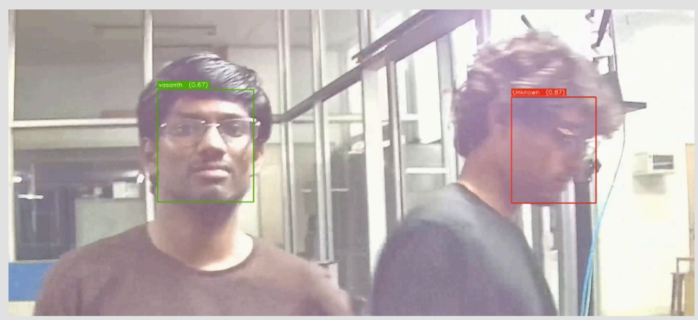
## Use cases and impact

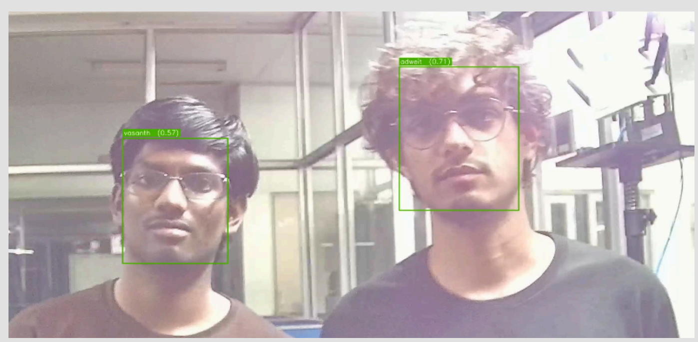
Use Cases:

- Institues
- Large lecture halls
- Corporate training
- Examination monitoring

Impact:

- Saves classroom time
- Eliminates proxy attendance
- Highly scalable
- Enables analytics





## Future Works

- Improve anti-spoofing techniques
- Real-time video processing instead of snapshots
- Student engagement tracking
- Integration with institutional ERP systems
- Edge AI optimization for faster processing